Security 03 | EAP-TLS Wired Policy

Overview

In this lab, we'll enhance the port security on our Raspberry Pi by enabling 802.1X authentication, configuring RadSec to communicate with AGNI, and verifying the wired EAP-TLS policy end-to-end.

What you'll learn:

- Enable 802.1X and MAC authentication on a wired port profile
- Configure RadSec on an EOS switch via Static Configuration Studio
- Create AGNI device groups, networks, and segments for wired EAP-TLS
- Verify wired dot1x sessions in AGNI and EOS CLI

Open Launchpad | **Login Help** | **Lab Topology**

Update Port Profile

Here we will update our existing Raspberry Pi port profile to enable 802.1X.

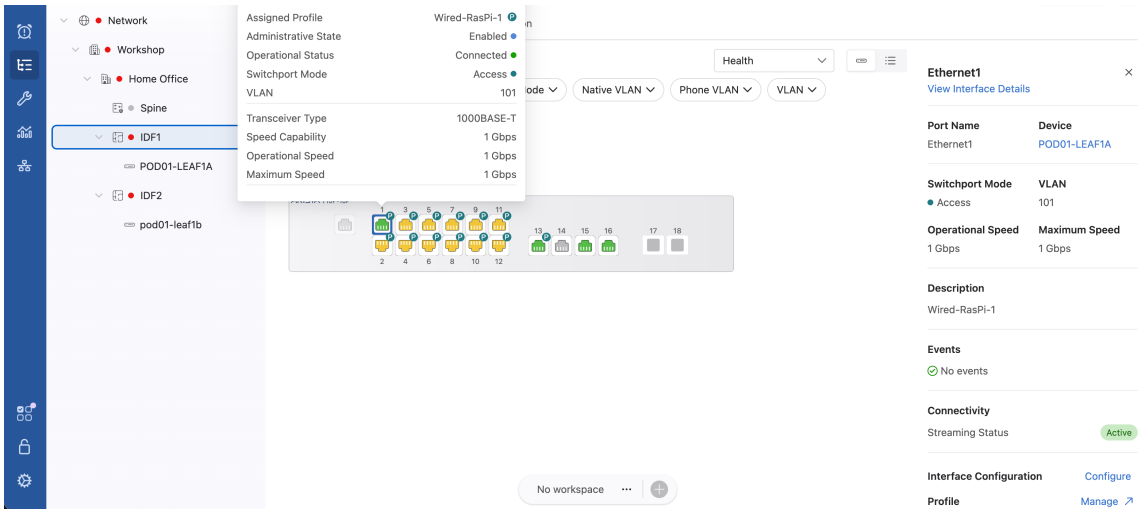
Single Workspace

You and your fellow student will work together to create the port profile for your campus fabric in a **single workspace**.

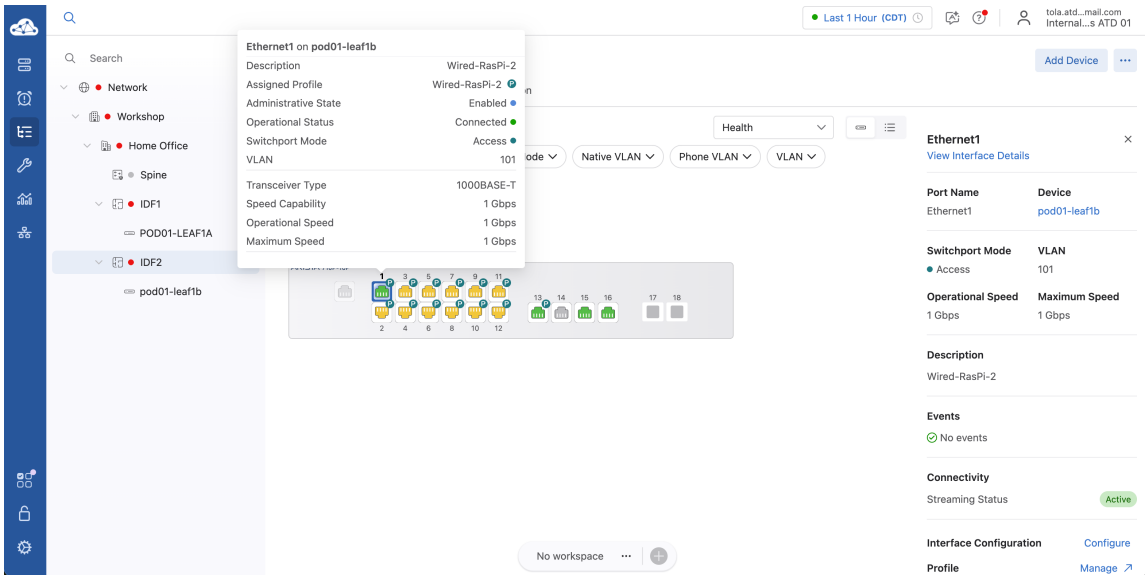
1. Let's update the `Wired-RasPi` port profile for our Raspberry Pi and enable 802.1X. Navigate to `Network Hierarchy` and select our `IDF`.
2. From the `Front Panel` view, verify that `Ethernet 1` is assigned to the `Wired-RasPi` profile from the Access Interface Configuration Guide.

Student 1

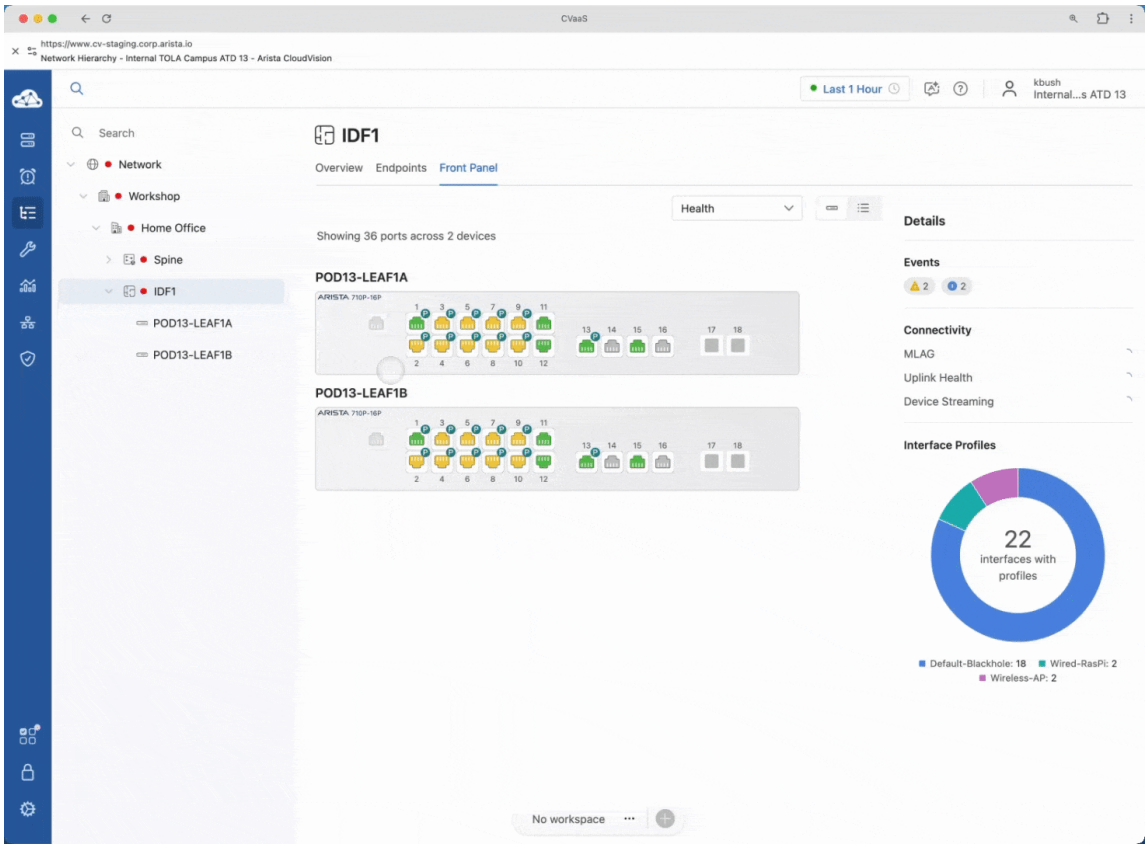




Student 2



3. Select the Raspberry Pi interfaces and navigate to configure a new port profile. Toggle `Parent Profile` and select the existing `Wired-RasPi` profile.

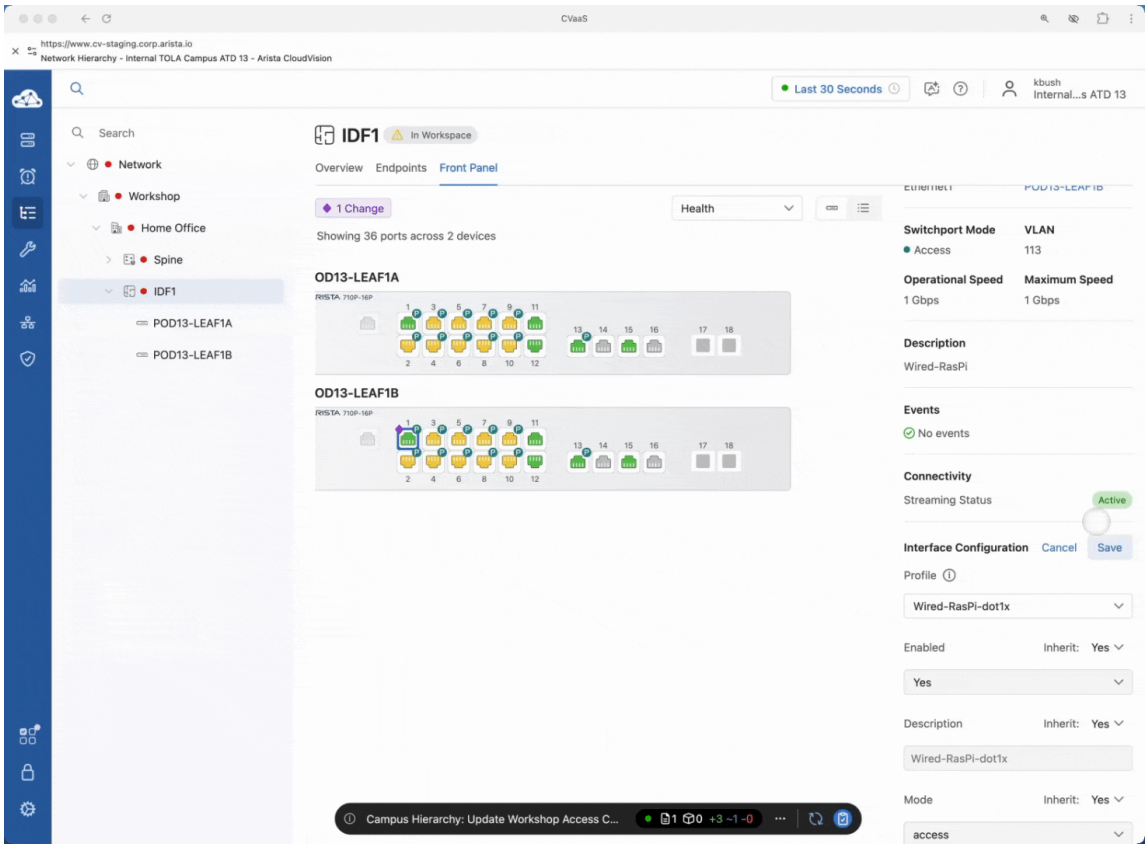


4. Override the default behaviors by toggling `Inherit from Parent Profile` and filling in with the following fields:

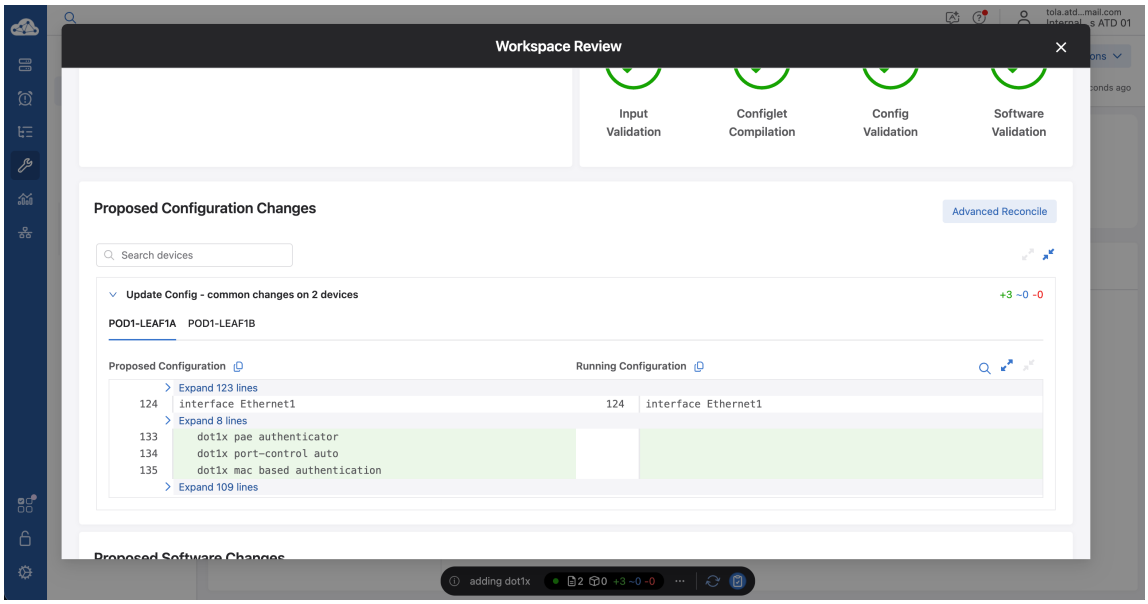
Wired-RasPi-dot1x

Section	Key	Value
General	(Student 1) Name	`Wired-RasPi-dot1x-1`
	(Student 2) Name	`Wired-RasPi-dot1x-2`
General	(Student 1) Description	`Wired-RasPi-dot1x-1`
	(Student 2) Description	`Wired-RasPi-dot1x-2`
802.1x General	Enabled	Yes
802.1x Mac-Based Authentication	Enabled	Yes

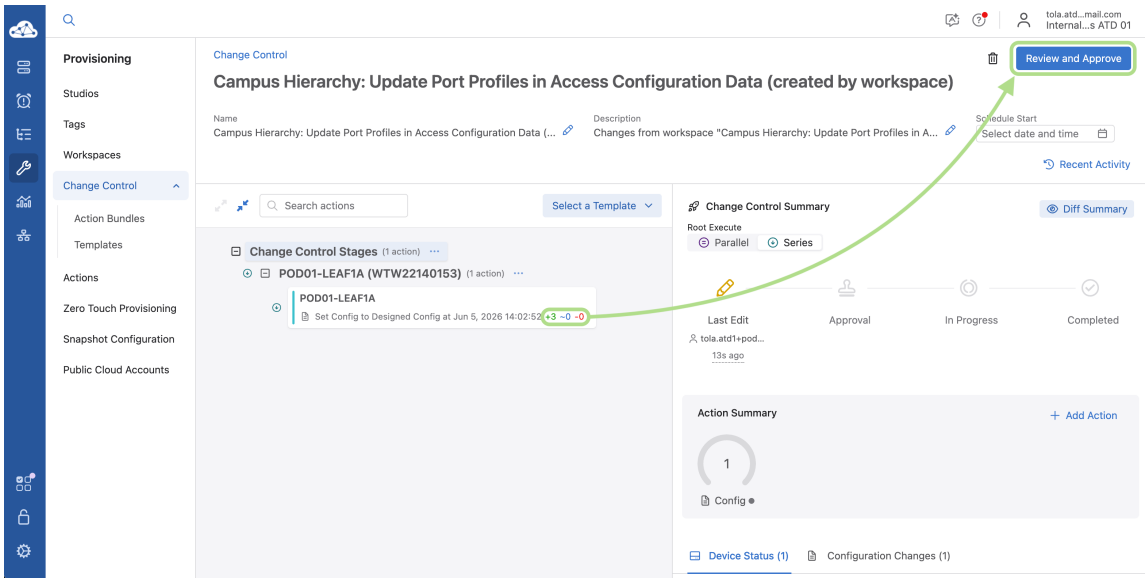
5. Select `Create` when you're done and we should trigger a new workspace. Select `Close` when that is finished.
6. Select `Save` under `Interface Configuration` and verify the change under `Review Workspace`.



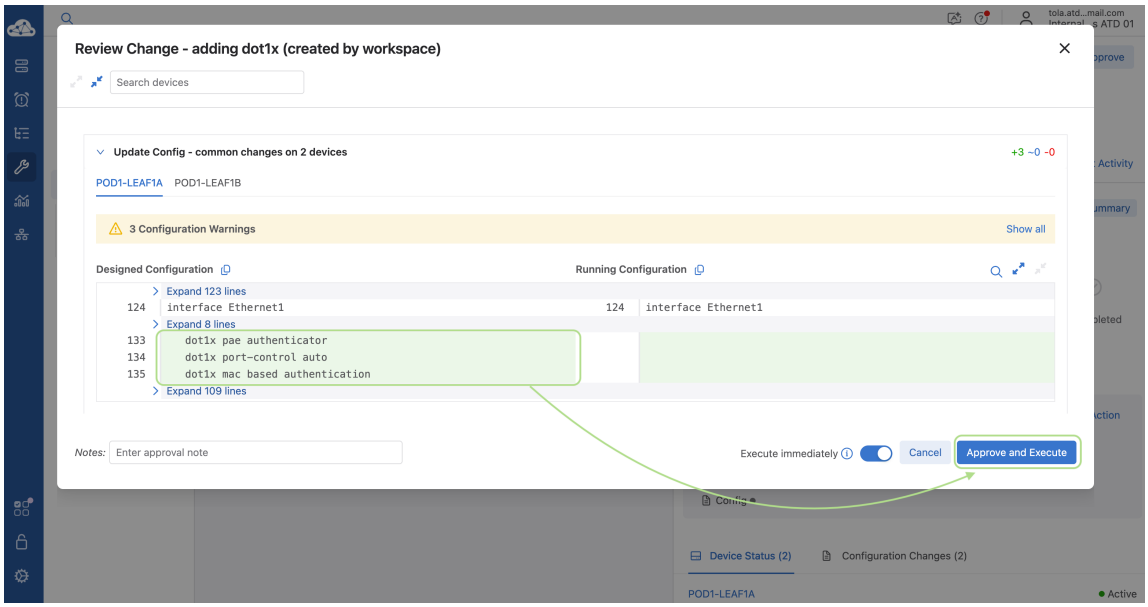
7. We can see the only studio changed is the `Access Interface Configuration`, and the ports assigned are updated.



8. Select `Submit Workspace` when you're ready and `View Change Control`.
9. Review the Change Control and select `Review and Approve`.



10. Toggle `Execute Immediately` and select `Approve and Execute`.

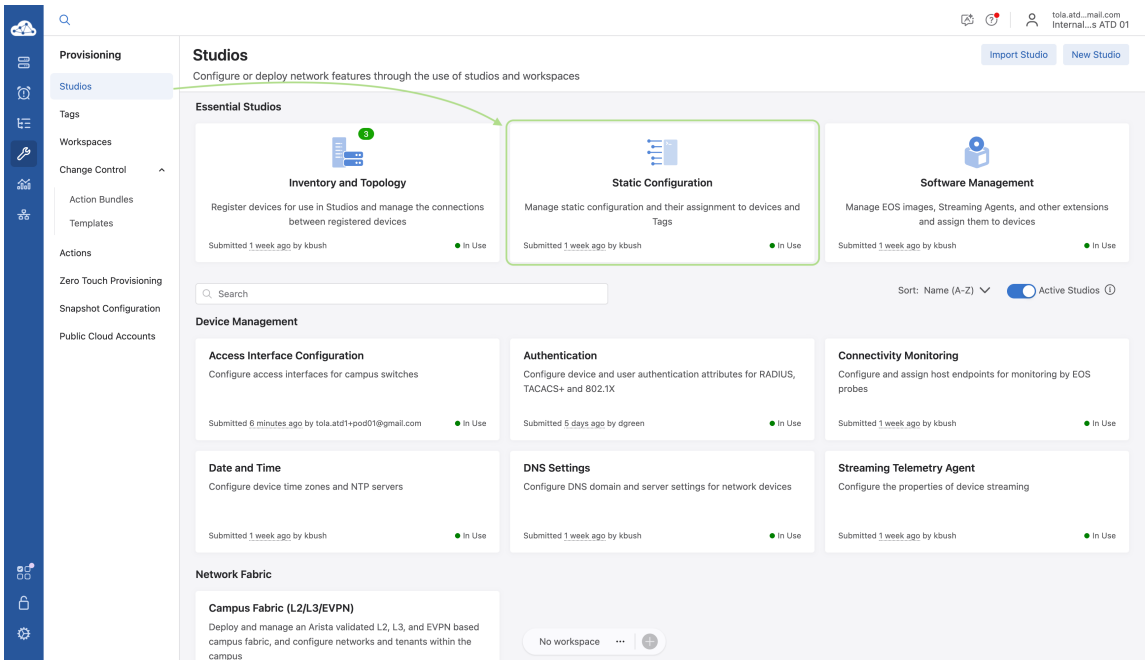


The port is now enabled for 802.1X. Let's now get your switch talking back to AGNI.

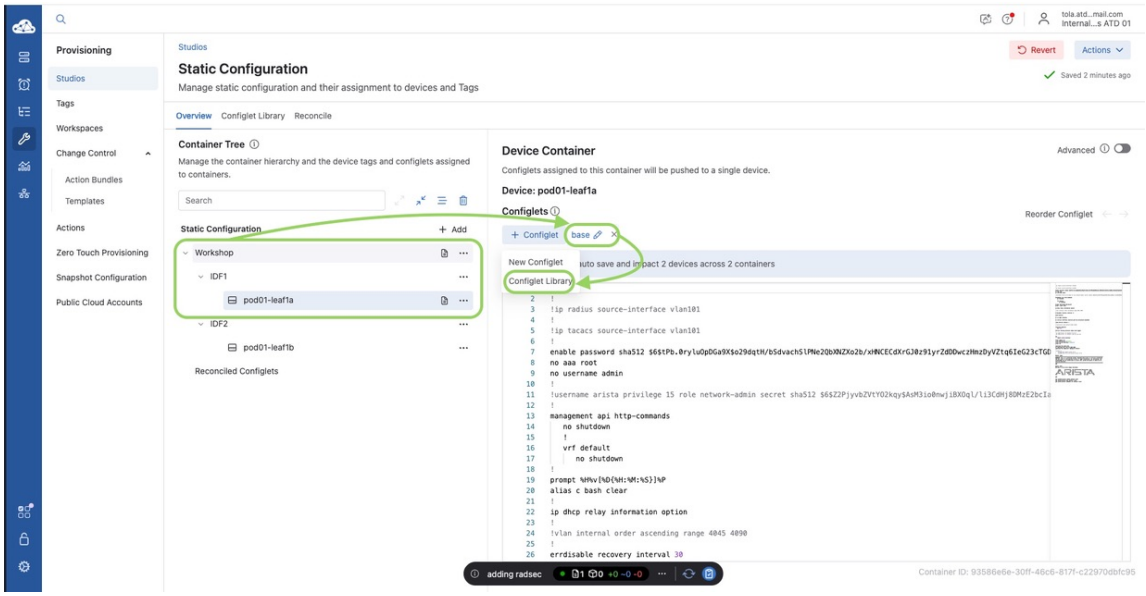
RadSec Spotlight

In this lab you will be configuring RadSec on your lab switches by adding the RadSec configuration to the switches via the `Static Configuration Studio`.

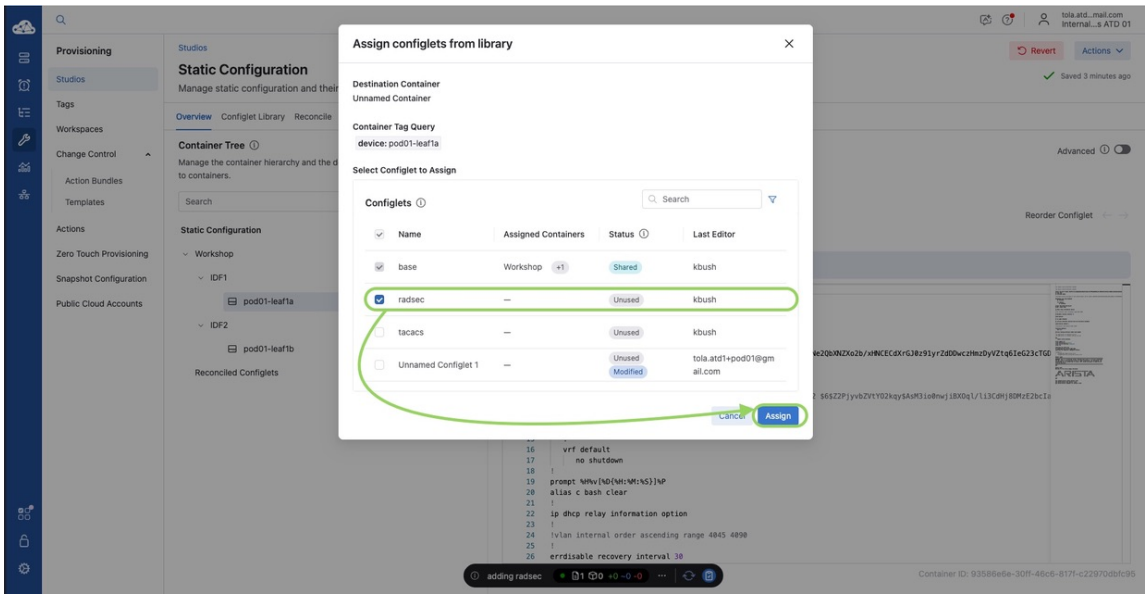
1. Select `Provisioning`, then select `Studios`.
2. Open the `Static Configuration Studio`.



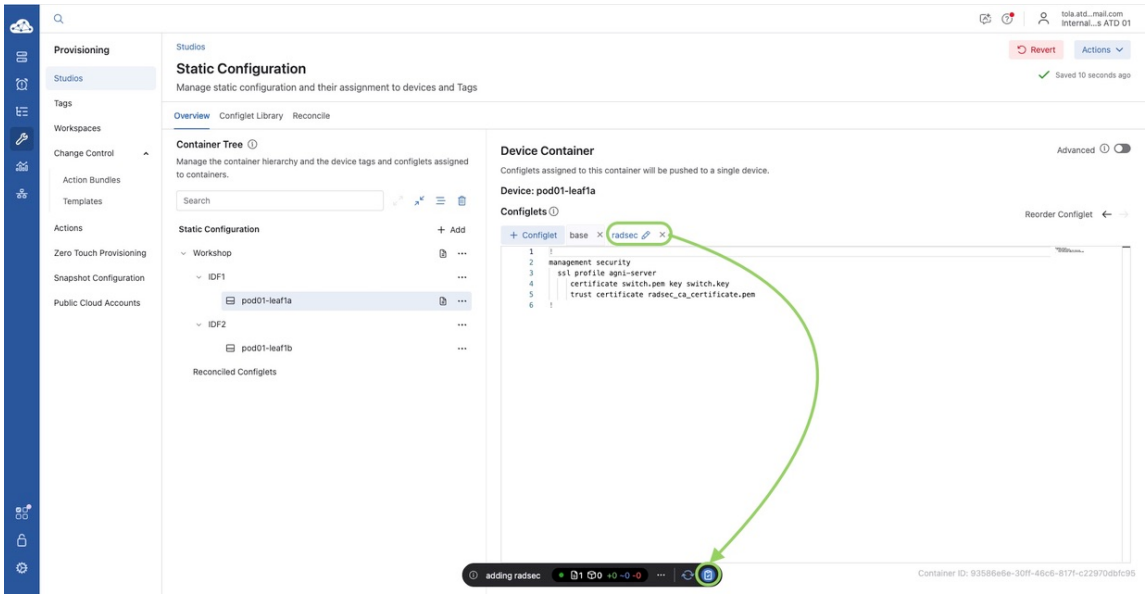
3. Create a new `Workspace`.
4. In the `Overview` window, select `Workshop>IDF(X)>Leaf1(X)`, then select `+ Configlet` followed by `Configlet Library`.



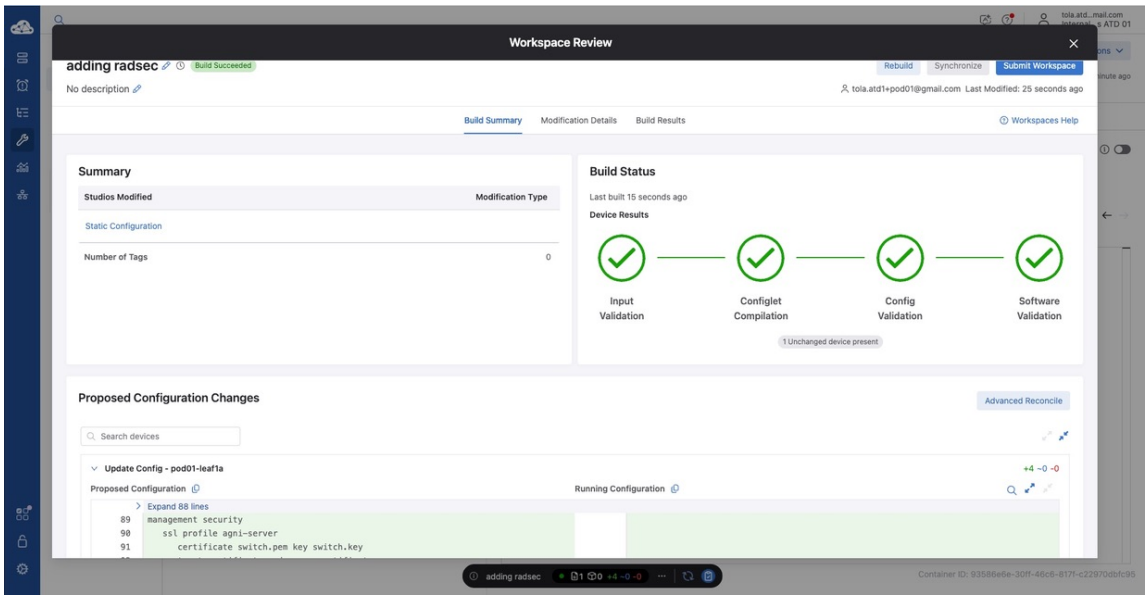
5. Select the configlet named `radsec` and select `Assign` to add the configlet to the `IDF` container.



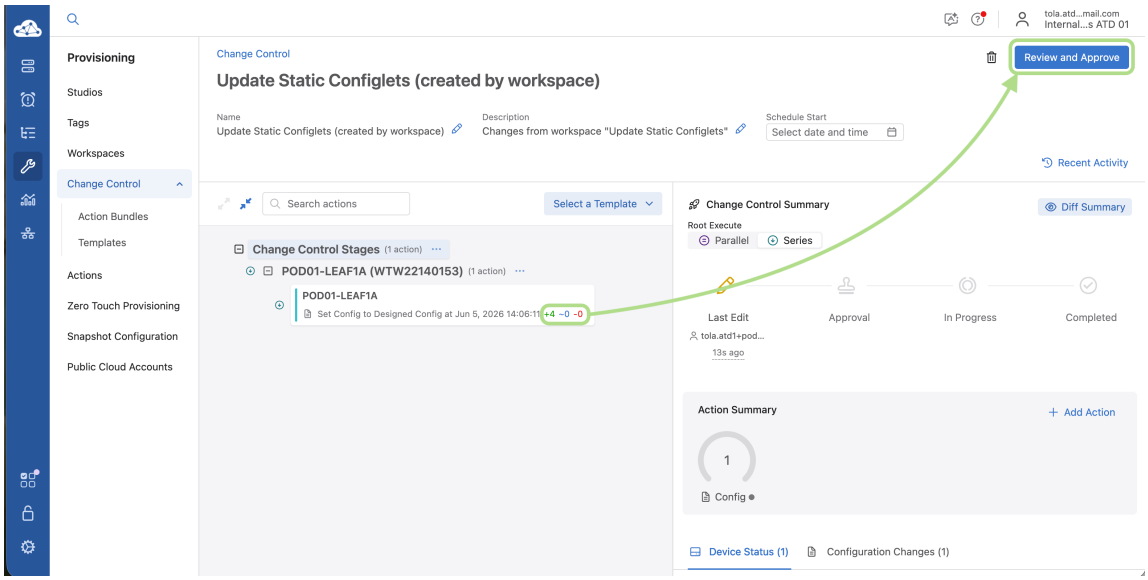
6. Select `Review Workspace` to review all the changes proposed to the CloudVision Studio.



7. Review the workspace details showing the summary of modified studios, the build status, and the proposed configuration changes for each device. When ready, select `Submit Workspace`.



8. Select `View Change Control` and review the Change Control. Select `Review and Approve` when ready.



9. Toggle `Execute Immediately` and select `Approve and Execute`.
10. The Change Control will execute and apply all the RadSec configuration changes to the device. This enables RadSec connectivity between the switch and AGNI.

What does this configuration do?

```
!
management security
    ssl profile agni-server (1)
        certificate pod00-leaf1a.crt key agni-private.key (2)
        trust certificate radsec_ca_certificate.pem (3)
!
```

(1) Create an SSL profile

(2) This is the switch key and certificate — generated on EOS, signed by AGNI, and installed in the store.

(3) This is the trusted certificate downloaded from AGNI and installed on the EOS certificate store.

Create Wired EAP-TLS Network and Segment

In this section we will create a Network and Segment in CloudVision AGNI to utilize a certificate-based TLS authentication method on a wired connection with a Raspberry Pi.

1. Navigate to `Access Devices > Devices` to confirm the RadSec connection is up.
2. Before configuring the network, we want to add each switch to an `Access Device Group`.

Access Device Group Settings

Field	Student 1	Student 2
Name	WIRED-A	WIRED-B
Description	WIRED-A	WIRED-B
Available Devices (`+ Add`)	POD##-LEAF1A	POD##-LEAF1B

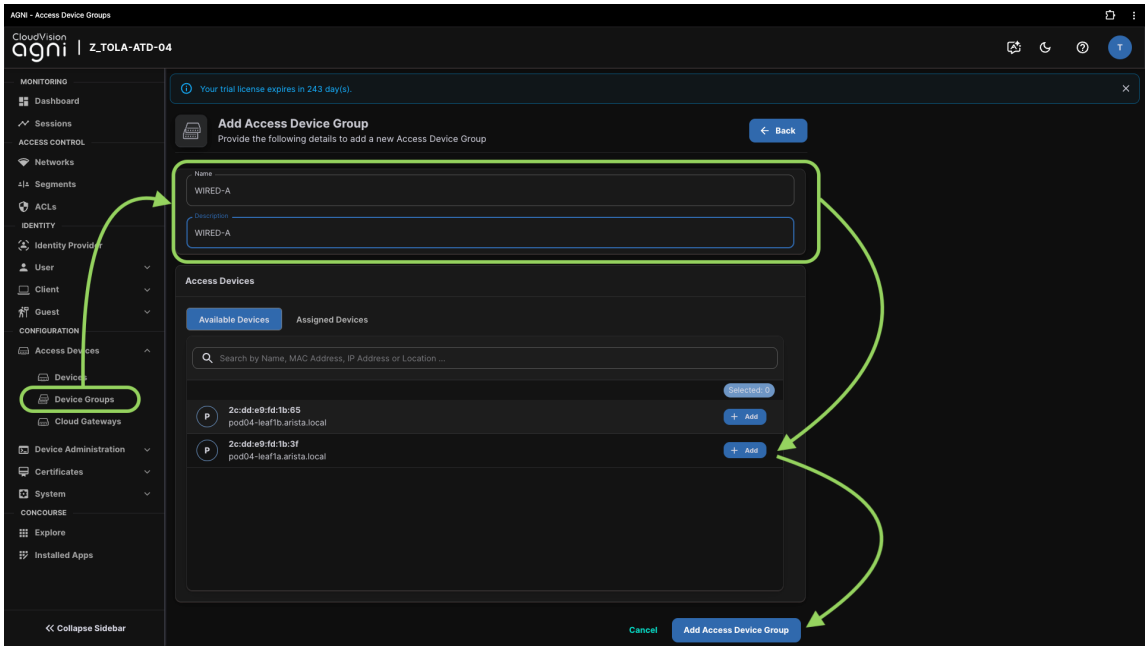
Student 1

a. Under `Access Devices`, select `Device Groups`.

b. Select `+ Add`.

c. Enter the `Name` and `Description`, then select `+ Add` for your switch.

d. Select `Add Access Device Group`.



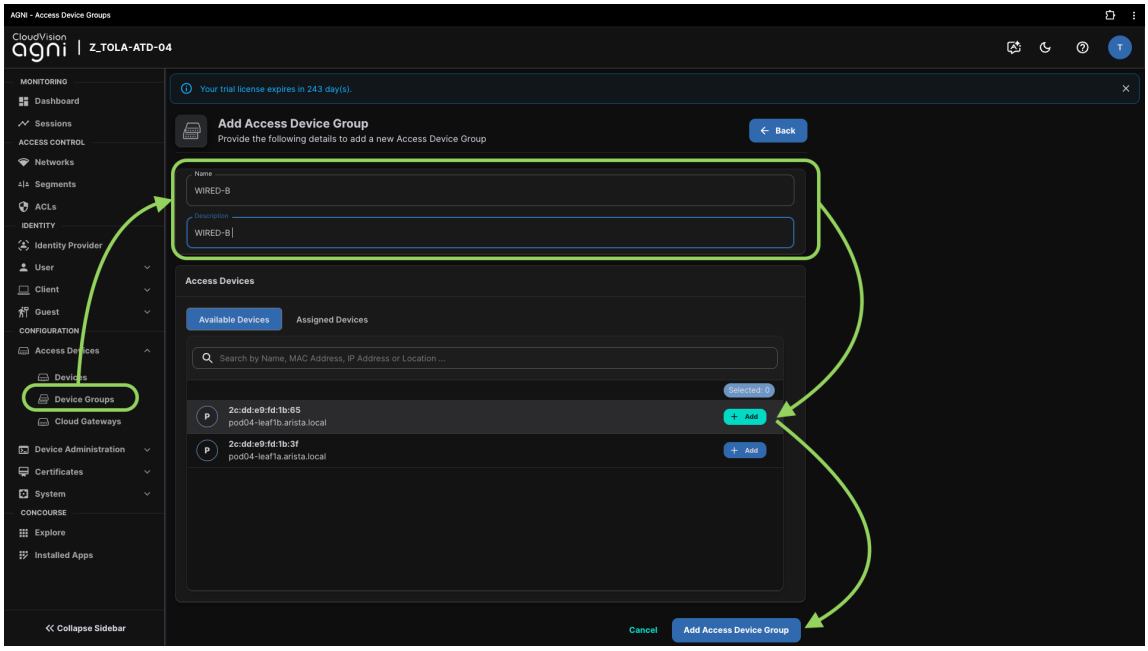
Student 2

a. Under `Access Devices`, select `Device Groups`.

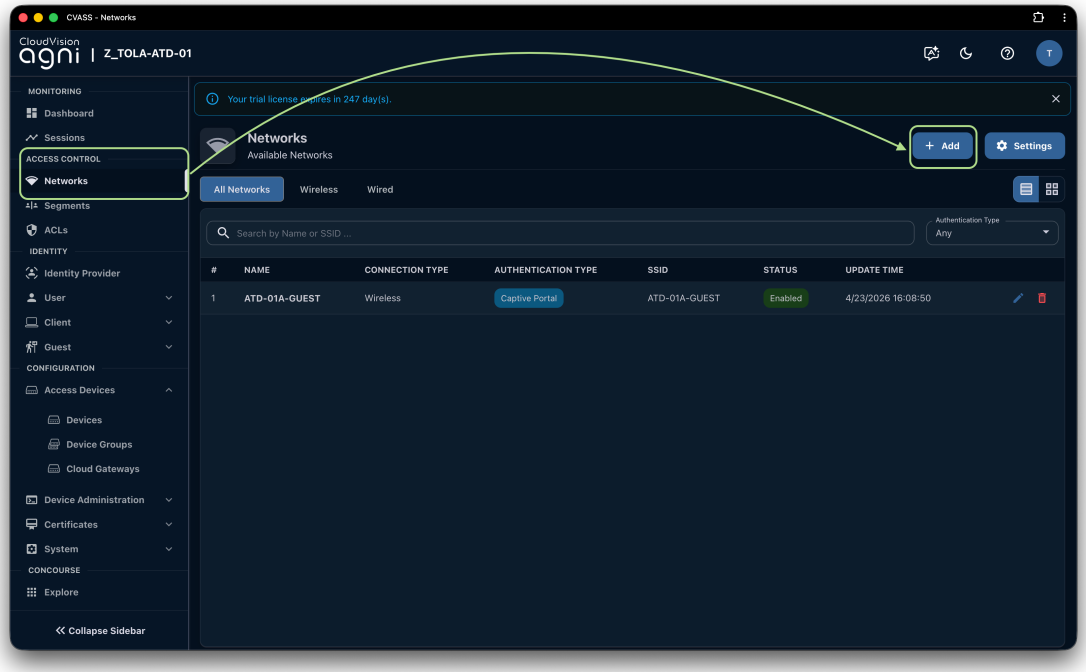
b. Select `+ Add`.

c. Enter the `Name` and `Description`, then select `+ Add` for your switch.

d. Select `Add Access Device Group`.



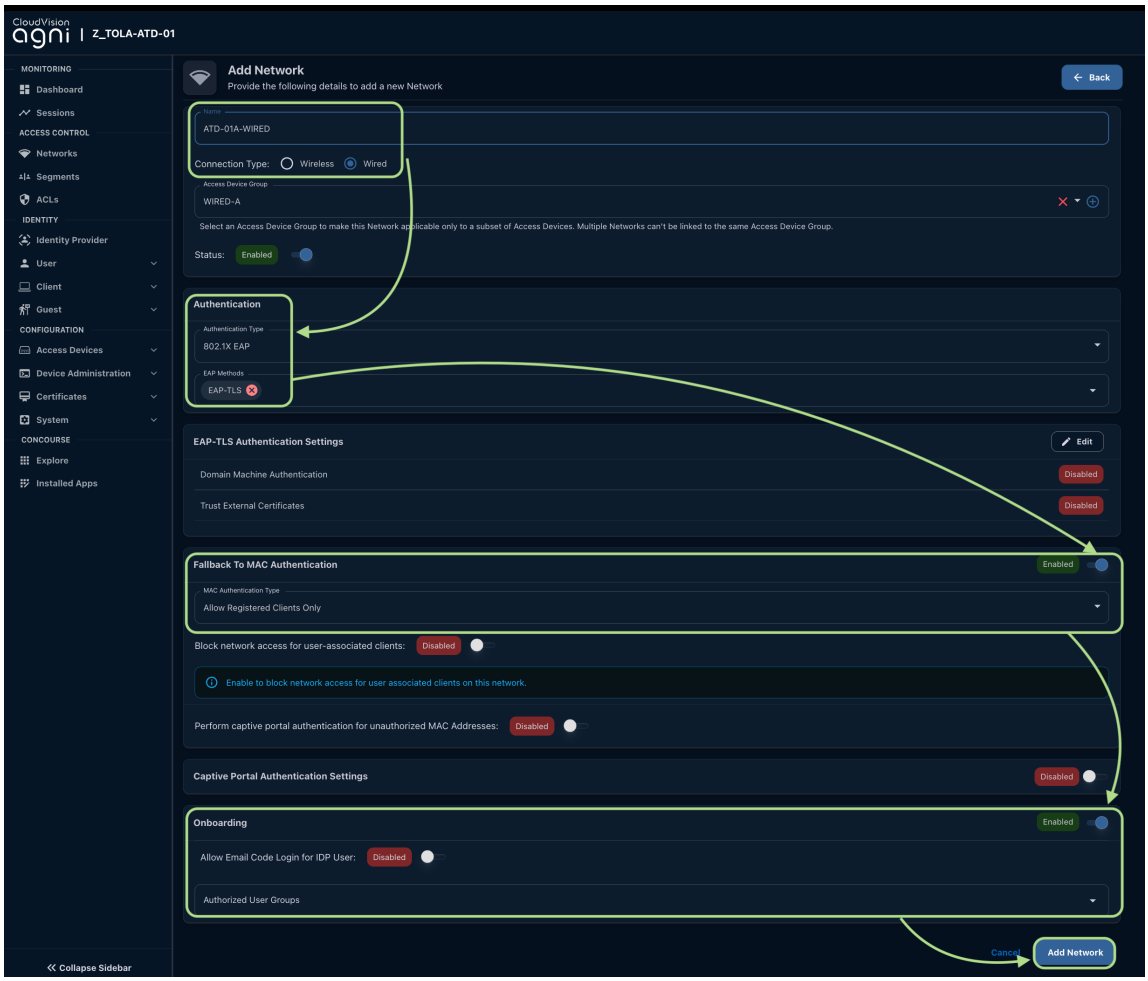
3. Select `Networks` and then select `+ Add`.



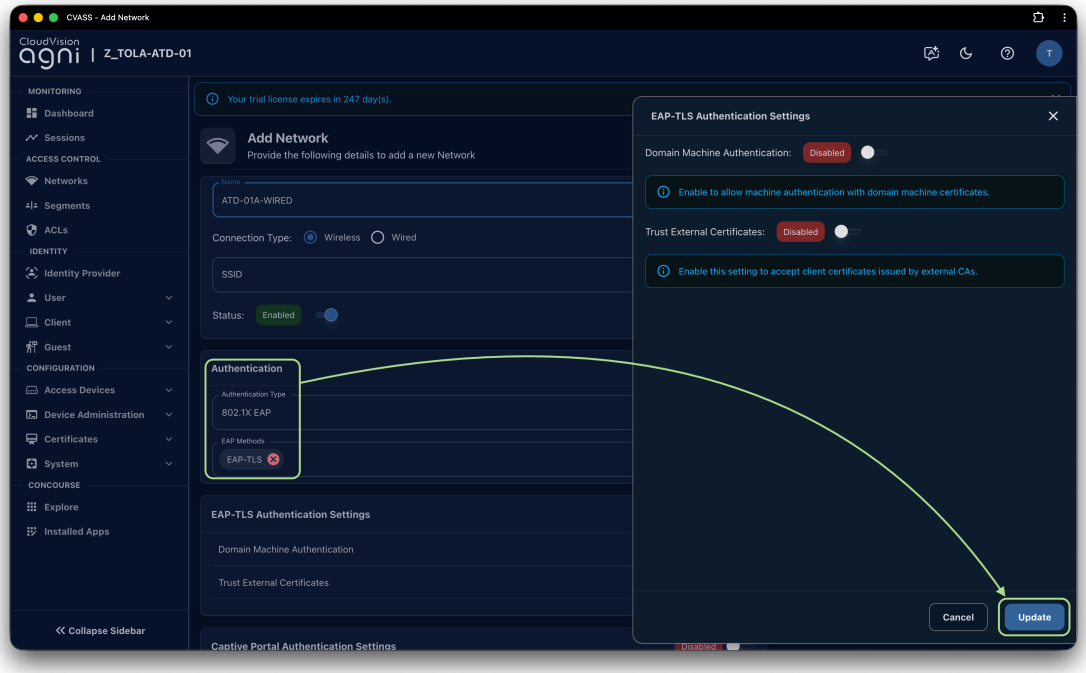
4. Fill in the following fields on the `Add Network` page. When selecting `EAP-TLS` under `EAP Methods`, leave `Domain Machine Authentication` and `Trust External Certificates` on `Disabled` and select `Update` at the bottom to resume Network Settings.

Network Settings

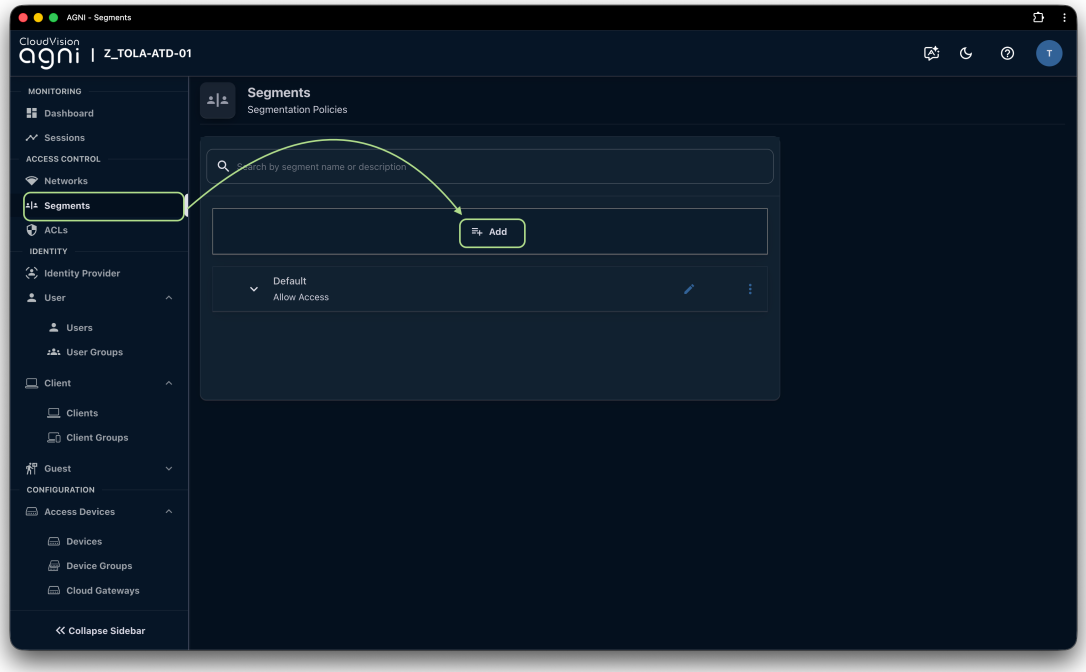
Field	Student 1	Student 2
Name	ATD-##A-WIRED	ATD-##B-WIRED
Connection Type	Wired	Wired
Access Device Group	WIRED-A	WIRED-B
Status	Enabled	Enabled
Authentication type	802.1X EAP (EAP Methods: EAP-TLS)	802.1X EAP (EAP Methods: EAP-TLS)
Fallback to mac Authentication	Enabled	Enabled
MAC Authentication Type	Allow Registered Clients Only	Allow Registered Clients Only
Onboarding	Enabled	Enabled
Authorized User Groups	Arista	Arista



5. When adding `802.1X EAP (EAP Methods: EAP-TLS)`, verify that `Domain Machine Authentication` and `Trust External Certificates` are disabled (should be by default) and select `Update` at the bottom.



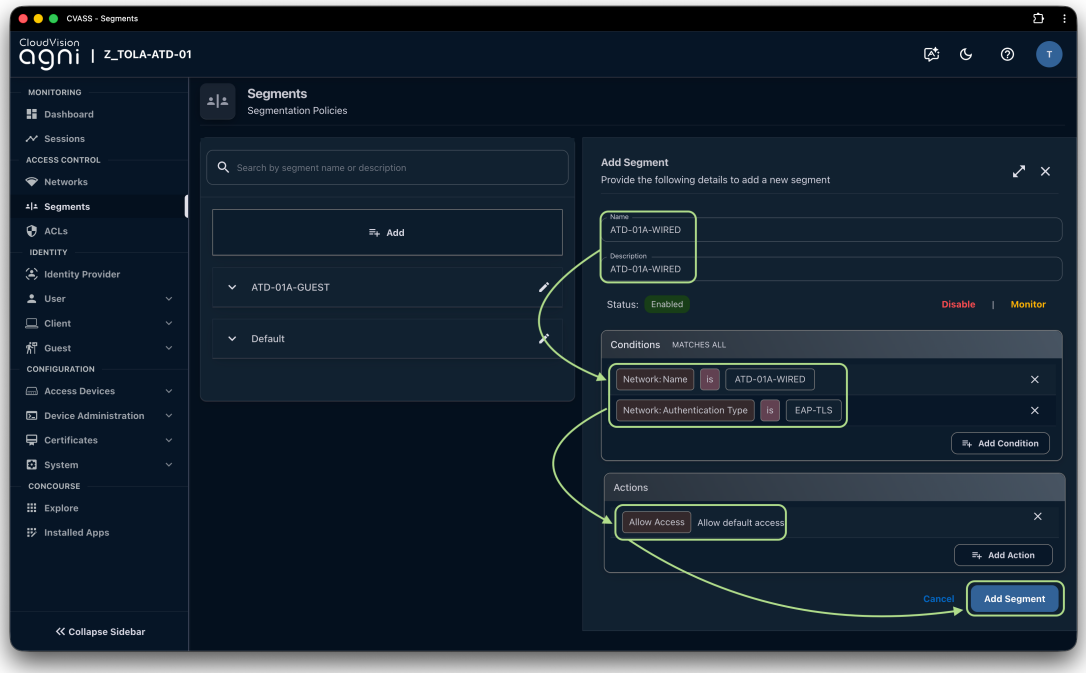
6. When done, select `Add Network` at the bottom of the screen.
7. Select `Segments` and then select `+ Add`.



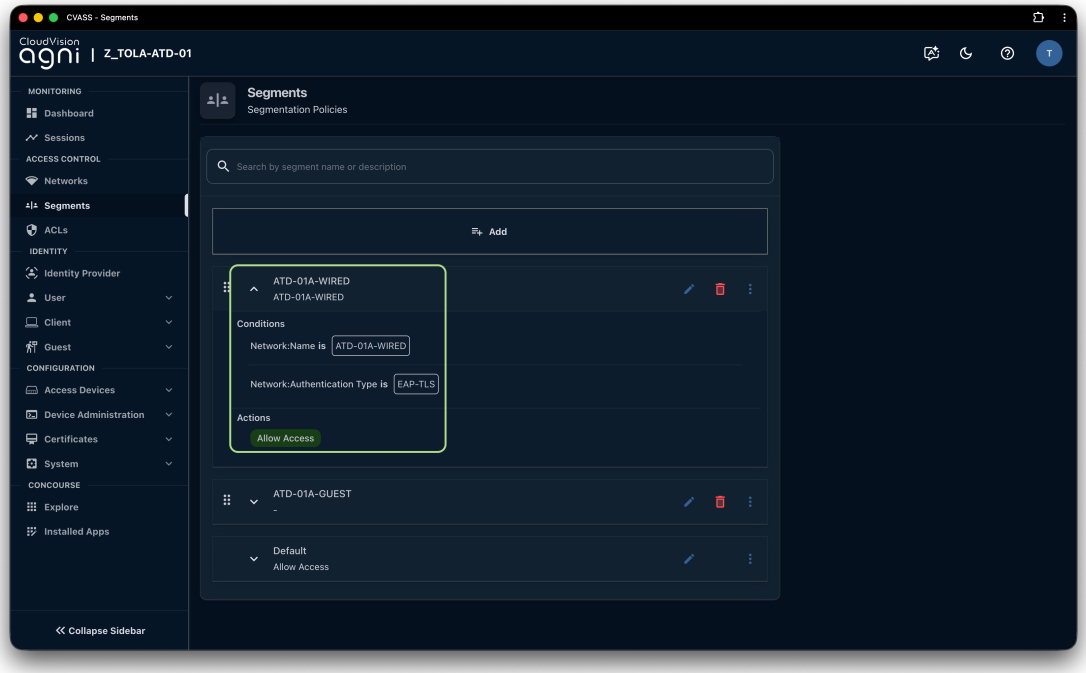
8. Configure the network segment with the following settings:

Segment Settings

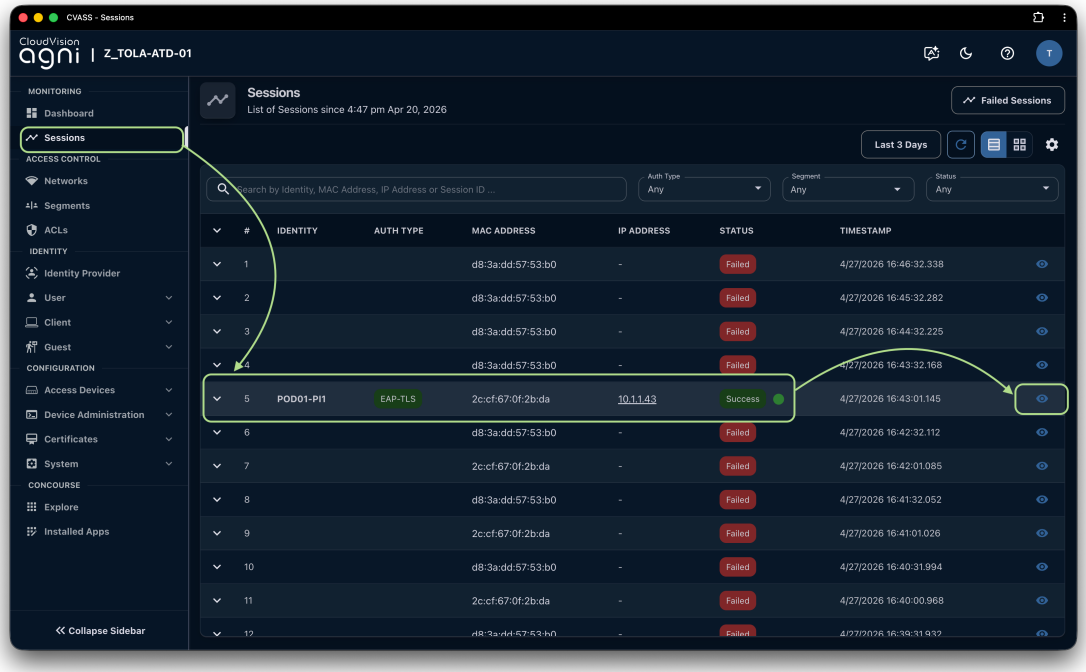
Field	Student 1	Student 2
Name	ATD-##A-WIRED	ATD-##B-WIRED
Description	ATD-##A-WIRED	ATD-##B-WIRED
Condition #1	`Network:Name is ATD-##A-WIRED`	`Network:Name is ATD-##B-WIRED`
Condition #2	`Network:Authentication EAP-TLS`	`Network:Authentication EAP-TLS`
Action #1	`Allow Access`	`Allow Access`

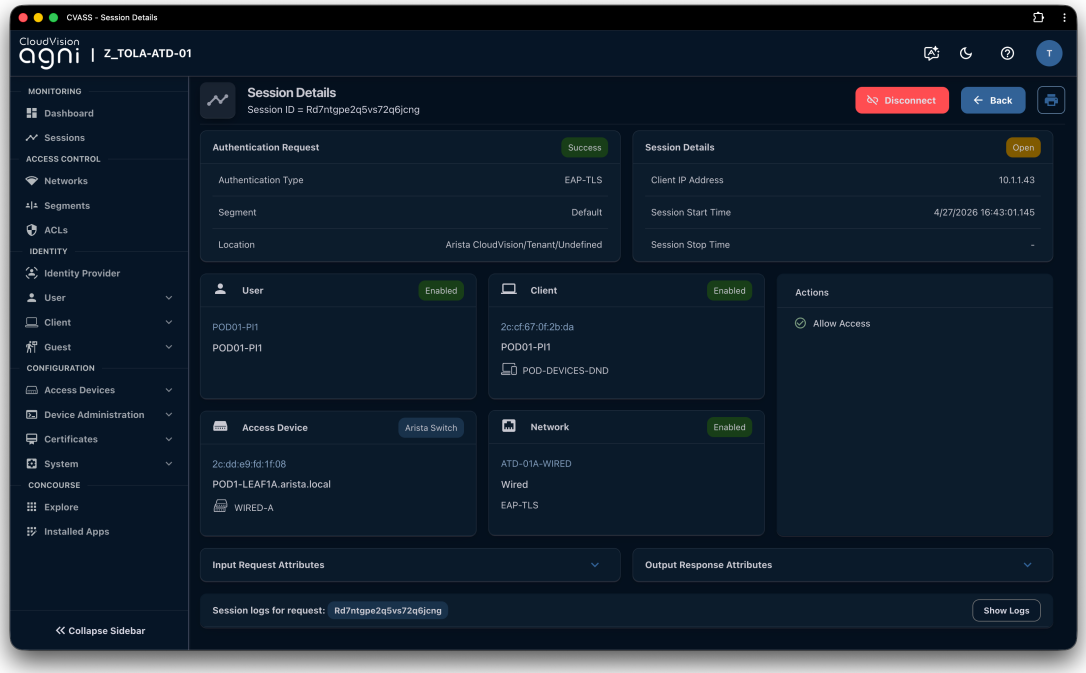


9. Select `Add Segment` at the bottom of the page.
10. You should now be able to expand and review your segment.



11. Select `Sessions` to see if your ATD Raspberry Pi has a connection via the wired connection.





Client Certificate

The Client Certificate has already been applied to the Raspberry Pi.

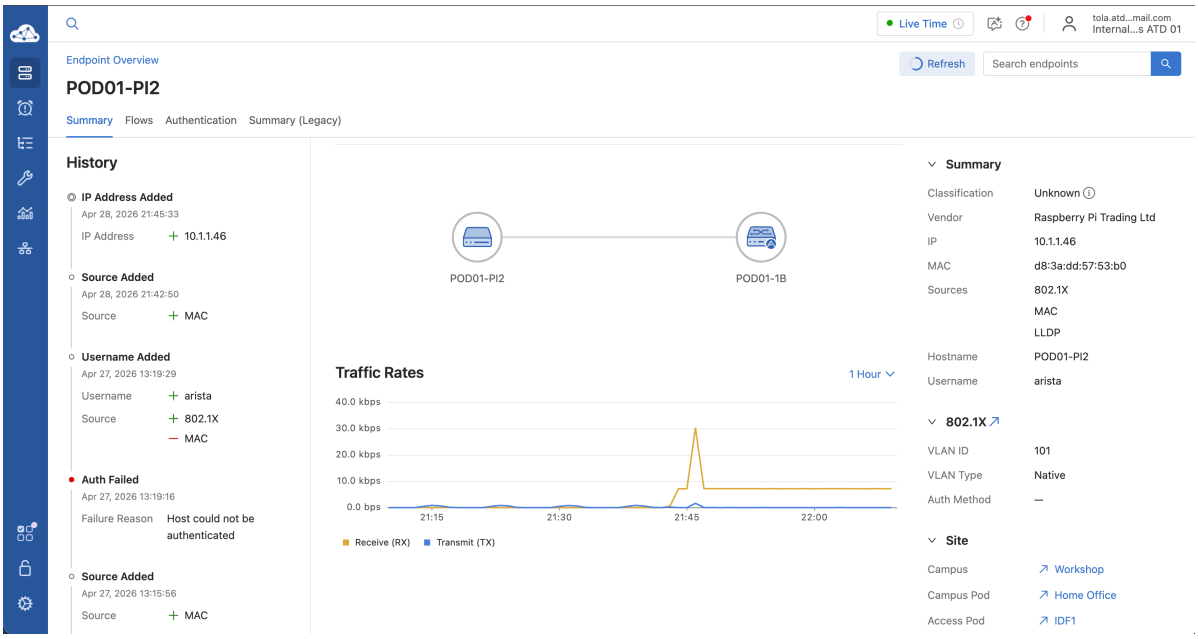
Validate and Verify Wired EAP-TLS Device

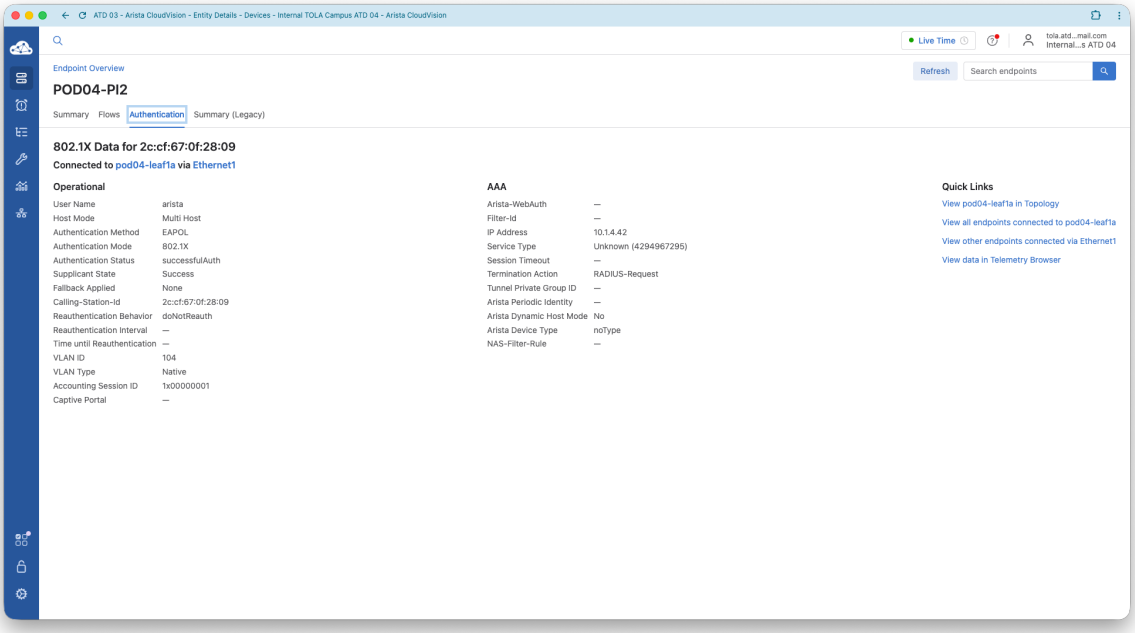
AGNI

1. Once the device is connected, you will be able to view the status of the connection and additional session details by selecting the eye icon to the right of the device.
2. AGNI will then display more in-depth session information regarding the device and connection.

CloudVision Endpoint Overview

Navigate to `Endpoint Overview` and search for a device on your pod. With sFlow enabled, you should be able to see more information about authentication, traffic flows, and a summary tab.





EOS CLI

You can also validate the session on the switch by issuing the following commands in the switch CLI.

```
show dot1x host
show dot1x host mac d83a.dd98.6183 detail
```

```
pod00-leaf1a#show dot1x host
Port      Supplicant MAC Auth  State                  Fallback
VLAN
--------- -------------- ----- ---------------------- ----------------------
----
Et2       d83a.dd98.6183 EAPOL SUCCESS                NONE

pod00-leaf1a#show dot1x host mac d83a.dd98.6183 detail
Operational:
Supplicant MAC: d83a.dd98.6183
User name: aristaatd01@outlook.com
Interface: Ethernet2
Authentication method: EAPOL
Supplicant state: SUCCESS
Fallback applied: NONE
Calling-Station-Id: D8-3A-DD-98-61-83
Reauthentication behaviour: DO-NOT-RE-AUTH
Reauthentication interval: 0 seconds
VLAN ID:
Accounting-Session-Id: 1x00000004
Captive portal:
AAA Server Returned:
Arista-WebAuth:
Class: Rcnlkerh9ci3s72u197e0|C4151a596-baab-444b-a4fd-ad40946d8b5f
Filter-Id:
Framed-IP-Address: 192.168.101.21 sourceArp
NAS-Filter-Rule:
Service-Type: None
Session-Timeout: 86400 seconds
Termination-Action: RADIUS-REQUEST
Tunnel-Private-GroupId:
Arista-PeriodicIdentity:
```

Additional References

- Configuring RadSec

🎉 **CONGRATS! You have completed the Security labs!** 🎉

2026-06-10